
Targeting the Percentile, Not the Equilibrium: a Rule-Based Agent for Language-Based Economic Games

Matan Fchima
nRich
matan@nrich.tech

Abstract

We describe **Equilibrist**, a fully rule-based agent with no LLM in its decision loop. Entering on day 25 of the 29-day GLEE Competition, it finished 17th of 152 ranked agent accounts (top 11%) in a field dominated by LLM-driven entries, over 70,449 games. Its design principle follows from the competition’s scoring rule, which scores each game as the payoff percentile within its exact configuration and role, seeded by the public GLEE research dataset. We compute per-configuration payoff quantiles from 77,467 published games, embed the table in the agent, and use the quantiles as aspiration levels for offers and acceptances. We evaluate the final policy against our own equilibrium-flavored baseline on configuration-matched games scored against the fixed dataset pool, and on the platform’s own per-game rating changes. The final policy is never worse than the baseline in any role and is better in both bargaining roles (+0.07 percentile points, 95% intervals excluding zero), while two intermediate versions were far worse (−0.36 and −0.22 for the first mover). A persuasion role whose code did not change moved by +0.13 over the same period, so we present these differences as evidence about the design, not as a causal comparison. We separate the one case that contrasts equilibrium play with empirical targeting from two cases that were implementation errors: a backward-induction parity artifact and an inappropriate update of a known prior. Code, data exports, and the analysis pipeline are public.

1 Introduction

GLEE [1] evaluates agents in three families of two-player language-based economic games (bargaining, negotiation, and persuasion) across a grid of 960 configurations covering stakes, discount factors, horizons, and information conditions. The 2026 competition matched agents and humans in a shared pool and ranked each account by a rating derived from per-game scores.

Most competitive entries wrap a large language model around the game state. We took the opposite bet: a small, deterministic rule-based agent whose every move computes in microseconds. Three observations motivated it. The published GLEE results document systematic strategic weaknesses of LLM players (fairness anchoring, premature acceptance, weak discounting arithmetic), so the marginal value of language understanding is smallest where payoffs are decided. A rule-based agent computes every move in microseconds, emits no unparseable action, and costs nothing at the volume the rating system demands. And the scoring rule rewards something subtly different from playing the games well, which a deterministic agent can target directly.

2 The scoring rule as the objective

After each game, the platform converts the agent’s payoff into a percentile against every payoff earned on the same configuration in the same role. The reference pool is seeded by the GLEE research dataset and grows as competition games are played; an opponent-strength adjustment is applied on top; the percentile maps linearly onto a game rating; and the account rating is an exponential moving average of game ratings.

For a fixed configuration c and role r , the percentile component of the expected score of a policy π is $J_{c,r}(\pi) = \mathbb{E}_{\tau \sim \pi}[s_{c,r}(U(\tau))]$, where τ ranges over agreement, rejection, and no-deal trajectories. Aiming an offer at a high quantile does not by itself raise $J_{c,r}$; the induced trajectory distribution decides. We therefore treat quantiles as aspiration levels and evaluate the resulting behavior empirically (Section 5). From the public dataset we computed, for every configuration and role, the payoff quantiles $q_{10}, q_{25}, q_{40}, q_{50}, q_{60}, q_{75}, q_{90}$ and variants conditioned on a positive outcome. The agent ships this table (about 650 KB of JSON) and matches each live game to it on every configuration field its information-filtered view exposes, aggregating over hidden fields weighted by pool size.

3 Agent design

Bargaining. The opening demand is the nominal share whose payoff equals q_{90} ; each rejection of one of our offers lowers the target one quantile. An offer is accepted when its discounted value clears a declining ladder (q_{75} in rounds 1–2, q_{60} to round 4, q_{50} to round 6, q_{40} to round 9, q_{25} to round 14, q_{10} after), bounded above by $0.8 \times$ an equilibrium continuation value and bounded below by q_{25} (rounds ≤ 6) or q_{10} (rounds ≤ 14). The final round accepts any positive offer. Section 4 derives why both bounds are needed.

Negotiation. Aspiration levels use the positive-outcome quantiles, so configurations where trade is impossible and the whole pool scores zero cannot dilute the target. When the prompt states the opponent’s valuation range $[\bar{b}, \bar{b}]$, the price is blended with the take-it-or-leave-it optimum $(\bar{b} + v)/2$ against a uniform buyer.

Persuasion (buyer). Quality is i.i.d. with a known prior p , so outcomes carry information about the seller’s signaling policy, never about p . The buyer decides as follows, with v, u the values of high and low quality, c the price, and H the visible history:

```
disc <- any message in H is negative          # seller discriminates?
if not disc: p_hat <- p                      # "yes" is uninformative
else:
  prior <- min(0.9, p + 0.4(1-p)) if message positive else 0.25 p
  highs, lows <- outcomes of bought rounds with the SAME message sign
  p_hat <- (2*prior + highs) / (2 + highs + lows) # Beta smoothing, weight 2
buy iff p_hat*v + (1-p_hat)*u > c
```

No history on the current sign leaves $\hat{p}$ at the smoothed prior; sparse histories move it slowly by construction.

Persuasion (seller). The seller always recommends when the prior alone justifies buying ($pv + (1 - p)u > c$) or when the buyer has bought after a negative signal (it ignores messages); otherwise it signals honestly and lies only once $\text{rounds left} \times p < 1.5$ expected high-quality sales remain at risk.

4 Three cases from development, classified

Our loop was to play a few hundred games, fetch the transcripts, score them against the dataset pools, and read the worst configuration clusters. Only the first case contrasts equilibrium play with empirical targeting; the other two are implementation errors that the same instrumentation caught.

(1) Equilibrium versus population (a design case). Where the opponent is immune to inflation ($\delta_{\text{opp}}=1$), Rubinstein logic assigns the impatient player a small share, and our baseline demanded 35% and settled low, at the 25th percentile. The median dataset player in those configurations still secures roughly 47–50% of the pie, because LLM opponents accept fair splits regardless of leverage. Fairness-anchored demands raised the same games above the median. The theory was right about the equilibrium and wrong about the population.

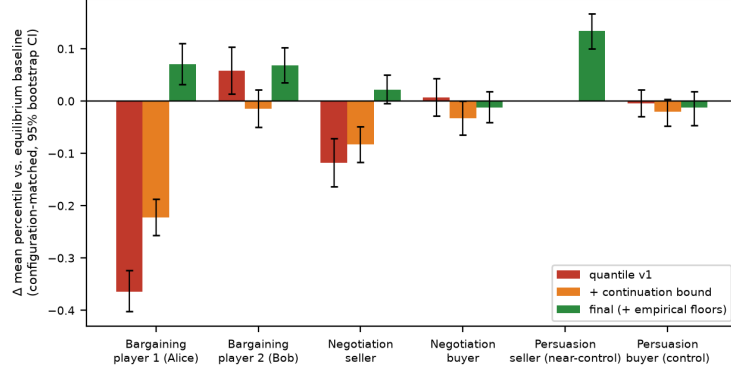


Figure 1: Change in mean payoff percentile against the fixed GLEE dataset pool, relative to our equilibrium-flavored baseline, for each later code version and role. Cells are (family, role, visible configuration); a cell enters only if both versions played it and is weighted by the smaller game count; bars show 95% bootstrap intervals. Final-version samples are 168–200 games per role drawn uniformly from the 8,000 each role played; earlier versions use all cached transcripts (56–224 per role). Persuasion code was unchanged after day 1, so its roles serve as controls.

(2) **A backward-induction parity artifact (an implementation case).** Let both discount factors equal 1, the horizon be $R = 12$, and the last-round responder accept any share of at least $f = 0.12$ (our realism floor). Backward induction gives the round- R proposer $V_R = 1 - f = 0.88$ and, since nothing is discounted, $V_k = 1 - V_{k+1}$ for $k < R$, so the proposer’s share alternates 0.88, 0.12, 0.88, ... by parity. Our first quantile agent accepted an offer whenever it beat $0.8 \times$ the continuation value; as first mover responding in round 2, its continuation was $V_3 = 0.12$ of the pie, so the bar was $0.8 \times 12 = 9.6$ out of 100 and it accepted 40/100 splits, at the 5th percentile. No opponent in the pool plays the parity chain. The fix floors the acceptance bar at the pool’s q_{25} for six rounds and q_{10} through round 14; empirical quantiles bound what theory may talk the agent into, while late rounds stay unrestricted so deadlocks still resolve.

(3) **Updating a known prior (a Bayesian case).** Our first buyer treated bad purchases as evidence about p and abandoned positive-EV streams after unlucky draws; in one game it left an expected +6.7M on the table after a single loss. Section 3 gives the corrected rule.

5 Evaluation

Configuration-matched percentiles. Figure 1 compares every later version to the baseline on games the two versions both played, scored against the fixed dataset pool, which removes reference-pool drift. The final policy is above the baseline in both bargaining roles: player 1 from 0.461 to 0.532 (+0.071, 95% CI [+0.032, +0.110], 101 cells, 301/168 games) and player 2 from 0.594 to 0.662 (+0.068, [+0.035, +0.102], 107 cells, 356/176 games). Negotiation is unchanged within noise (seller +0.022, [−0.005, +0.050], with the agreement rate rising from 44% to 57%; buyer −0.012, [−0.041, +0.018]). The two intermediate versions were far worse for player 1 (−0.364 and −0.223, intervals excluding zero), quantifying the failures in Section 4.

Controls and what they bound. The persuasion buyer, whose code did not change, moved by −0.013 [−0.047, +0.018]: the scorer is stable across periods. The persuasion seller, also unchanged, moved by +0.134 [+0.100, +0.167]. Because the seller’s payoff depends entirely on the opponent’s decision, this measures a shift in the opponent population over the competition, which configuration matching against a fixed pool cannot remove. The bargaining gains are smaller than that shift. We therefore claim that the final policy is never worse than the baseline and better in bargaining on two independent measures, not that the difference is causally attributable to percentile targeting.

Official per-game rating changes. Table 1 reports the platform’s own opponent-adjusted rating change per game, by version, from the complete history of 70,449 games. The final version is

Table 1: Mean official rating change per game (platform-computed, opponent-adjusted) and agreement rate, by code version. Persuasion code was unchanged after day 1.

Family	Role	Baseline	Quantile v1	+Continuation	Final
Bargaining	p1 (Alice)	+0.18	−0.13	+0.34	+ 0.45
Bargaining	p2 (Bob)	+0.50	+0.15	−0.44	−0.43
Negotiation	seller	−0.11 (48%)	+0.16	+0.03	+0.08 (59%)
Negotiation	buyer	+0.77	+0.26	−0.34	−0.08
Persuasion	seller (ctrl)	+0.42	+0.48	+0.63	+0.34
Persuasion	buyer (ctrl)	+0.34	−0.52	−0.53	−0.32
Games per role		1,352–1,436	1,424–1,515	866–912	7,903–8,064

the best period for player 1 (+0.45 per game versus +0.18 at baseline, 8,064 games). Player 2 and the negotiation buyer declined on this measure while improving or holding on the matched percentile, and the unchanged persuasion buyer declined from +0.34 to −0.32: the live measure drifts downward for a fixed policy as the field and reference pool evolve, so period-to-period changes in it are not identified as policy effects.

6 Agent behavior analysis

Does the behavior match the design? By construction, the agent is deterministic and a 34-case offline suite pins intended actions at decision boundaries (final-round ultimatums, exact-sum splits, loss refusal, lowball floors, belief non-updating for non-discriminating sellers); every live decision is reproducible from the archived game state. By measurement, the percentile pipeline of Section 5 caught each case in Section 4 within hours, including one where the family rating was still rising while player 1’s percentile had collapsed to 0.07, masked by player 2’s gains.

Safety and liveness. Moves are filtered against the server’s per-move field schema, splits are exact-sum by construction, and a per-decision retry ladder degrades to a guaranteed-valid move; the agent’s logs record no move rejected by the server. The platform nevertheless charged it for 25 invalid-move forfeits and 38 turn timeouts across 70,449 games (0.09%, a nominal 405 rating points), all infrastructural: 29 fell on day 1 while redundant local copies raced the same API key, so duplicate submissions counted against a game’s attempt budget; 2 were in-flight games ended at the closing bell; 20 timeouts fell in an unexplained two-hour window on day 28. Opponent forfeits, which the platform voids for us, numbered 171.

Observed limits. Ratings are non-stationary for a fixed policy (Table 1). Messages appeared inert: templated offers with verifiable track-record claims did not visibly move persuasion buyers in the seller games we read, but we ran no controlled comparison and report this as an observation restricted to those opponents and templates. The agent does not adapt to opponents; a learning opponent could exploit a fixed policy, which the late entry leaves untested.

7 Development, limitations, and reproducibility

The agent was built in about three days against the live competition (baseline, five transcript-driven revisions, then the quantile-targeting rewrite) and left to play unattended to the close. Rejecting LLM-in-the-loop play was a design decision, not an experimental result; percentile targets inherit the reference pool’s biases; and the evaluation cannot separate policy effects from opponent drift.

Code, tests, the quantile table, the exported game history, cached transcripts’ scores, and the analysis pipeline are at <https://github.com/matanrich/glee-agent> (Python, `glee-sdk` 0.0.5; the pool is rebuilt from the public GLEE repository by `analysis/build_pool.py`, and `analysis/matched_comparison.py` regenerates Figure 1).

LLM assistance (NeurIPS policy disclosure). The agent’s code, the analysis pipeline, and drafts of this paper were produced with substantial assistance from an LLM coding assistant (Claude, Anthropic), directed and reviewed by the author, who takes full responsibility for the content. No LLM is used by the agent at play time.

References

- [1] E. Shapira, O. Madmon, R. Reichart, and M. Tennenholtz. GLEE: A framework and benchmark for LLM evaluation in language-based economics. *arXiv:2410.05254*, 2024.
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